

CONTACT INFORMATION

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Citizenship: Germany

EDUCATION

- 2018** **Ph.D., Astronomy and Astrophysics (GPA: 4.0 US-scale, 18.08.2018)**
Steward Observatory, The University of Arizona, Arizona, AZ, USA
Thesis title: "The Formation and Evolution of Supermassive Black Holes: The Extremely Luminous Quasar Survey"
Advisor: Dr. Xiaohui Fan
- 2014** **M.Sc., Physics (Grade: 1.0, 01.09.2014)**
Christian-Albrechts-Universität zu Kiel, Kiel, Germany
Thesis title: "Evolution of Supermassive Black Holes and their host galaxies"
Advisor: Dr. Xiaohui Fan, Prof. Dr. Wolfgang J. Duschl
- 2011** **B.Sc., Physics (Grade: 1.7, 19.10.2011)**
Christian-Albrechts-Universität zu Kiel, Kiel, Germany
Thesis title: "Temporal evolution of self-gravitating accretion discs: self-similar solutions?"
Advisor: Prof. Dr. Wolfgang J. Duschl

PROFESSIONAL APPOINTMENTS

- since 2023 **Emmy Noether Junior Research Group Leader**
Hamburg University, Hamburg, Germany
- 2021 — 2023 **Postdoctoral Researcher** in the group of Dr. Joseph Hennawi
Leiden University, Leiden, The Netherlands
- 2018 — 2021 **Postdoctoral Researcher** in the groups of Dr. Eduardo Bañados and Dr. Fabian Walter
Max Planck Institute for Astronomy, Heidelberg, Germany
- 2012 — 2018 **Graduate research and teaching assistant**
Steward Observatory, University of Arizona, USA

PUBLICATIONS IN REFEREED JOURNALS

76 publications since 2015, with a total of > 5200 citations and an h-index of 35.
(11 as first author, 3 as second author led by an advised student/postdoc)
A complete publication list is enclosed and available on [NASA/ADS](https://nasa.ads).

GRANTS & THIRD PARTY FUNDING

- 2026 **Verbundforschung Astronomie & Astrophysik (pre-approved , 2-year postdoc position)**
Deutsches Zentrum für Luft- und Raumfahrt (DLR)
- 2026 **Excellence Cluster: Quantum Universe (Two 3-year PhD positions)**
Deutsche Forschungsgemeinschaft (DFG)
- 2025 **Verbundforschung Astronomie & Astrophysik (179,553 €)**
Deutsches Zentrum für Luft- und Raumfahrt (DLR)
- 2022 **Emmy Noether Independent Junior Research Group (1,750,101 €)**
Deutsche Forschungsgemeinschaft (DFG)
- 2015 **Science PI: HST, Cycle 23, Archival Proposal 14312 (\$149,266)**
Space Telescope Science Institute

SCIENTIFIC LEADERSHIP

- since 2025 Appointed **Co-lead** of Euclid work package QSOs, Euclid Consortium
- since 2025 Appointed **Co-lead** of Euclid DR1 Key Project "Discovery of high redshift Euclid quasars", Euclid Consortium
- since 2025 **Chair** of the "SMBH and AGN" working group within the COSMOS-3D collaboration (60 people)

AWARDS, HONORS, AND SCHOLARSHIPS

- 2026 **QU Best Paper Award (YIG leader)**
Excellence Cluster Quantum Universe
- 2019 **Best Poster Prize**
IAU Symposium 352, Portugal, June 2019
- 2012 **Fulbright Scholarship**, University of Arizona, 2012–2013
The German-American Fulbright Commission
- 2011 **Annual Award for excellent academic achievements and commitment**
Association for the support of the Faculty of Mathematics and Natural Sciences, Kiel University
- 2008 **Student Scholarship**, 2008–2014
German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes)

APPROVED TELESCOPE PROPOSALS

- Since 2018 PI on 7 successful observing proposals across 4 facilities (JWST, IRAM/NOEMA, LBT/LUCI, Gemini/GNIRS), including 2 JWST GO programs.

Proposals as Principal Investigator (since 2018)

- 2026 **JWST** (Survey; GO 11173)
"Pushing the Edge: A JWST Survey to Discover the Faintest Euclid QSOs at Cosmic Dawn"
- 2024 **JWST** (12 h; GO 5734)
"Placing a $z=7.27$ AGN in context with cosmic structure formation"
- 2021 **IRAM/NOEMA** (13 h)
"NOEMA's unobscured view on the host galaxies of a $z \sim 3.3$ binary quasar"
- 2020 **IRAM/NOEMA** (50 h)
"Unveiling the host galaxies of the most luminous quasars at $z \sim 3.5$ "
- 2019 **IRAM/NOEMA** (4 h)
"Unveiling the host galaxy of an ultra-luminous super-Eddington quasar at $z = 3.7$ "
- 2018 — 2020 **LBT/LUCI** (14 h)
"Weighing the most massive accreting black holes at Cosmic Noon"
- 2018 — 2020 **Gemini/GNIRS** (15 h)
"Weighing the most massive accreting black holes at cosmic noon"

Selected Proposals as Co-Investigator (since 2025)

- 2026 **JWST Cycle 5**
GO 10464 (PI: Davies, 72 h): "Echoes of the First Quasars"
GO 9841 (PI: Wang, 58 h): "Growing the most distant radio-loud quasars in the structure formation paradigm"
GO 9483 (PI: Tanaka, 18 h): "Spectroscopic Confirmation of the First Robust $z \sim 10$ LRD Candidate"
- 2025 **JWST Cycle 4**
GO 9180 (PI: Hennawi, 94 h): "Ushering in the JWST Era of Precision Constraints on Reionization"
6827 (PI: Eilers, 110 h): "From Dawn till Noon: Chronicling the Cosmic History of Black Hole Growth"
- HST Cycle 33**

GO 18112 (PI: Wolf, SNAP): "Beyond the Quasar Redshift Frontier: Uncovering Rapidly Accreting Supermassive Black Holes at $z > 8$ with HST/WFC3 and Euclid"

SUPERVISION EXPERIENCE AND TRAINING

Training

2025 Reflect Upon Your Leadership Role (CUI-Training, University of Hamburg)
 2024 **Supervisor Workshop Part 1** (MINGZ, University of Hamburg)

Primary Supervision

3 PhD students **Tatsuyuki Sekine** (since 2025, DAAD fellow, University of Hamburg), **Katharina Jurk** (since 2025, University of Hamburg), **Radha Anil Gharapurkar** (since 2024, University of Hamburg)
 2 Master students **Violet Moore** (since 2026, University of Hamburg, w/ F. Guarneri), **Arthur Lawrenz** (2025–2026, University of Hamburg, w/ F. Guarneri))
 3 Bachelor students **Lea Trahmer** (2025–2026, University of Hamburg, w/ R. Gharapurkar), **Niklas Knop** (2024, University of Hamburg, w/ F. Guarneri), **Lukas Wenzl** (2018–2019, MPA, 1 publication)

Co-Supervision

1 PhD student **Ben Wang** (2022–2026, Leiden Observatory)
 1 Master student **Daming Yang** (2021–2022, Leiden Observatory, 1 publication)

TEACHING

2026 **Full Lecture (co-taught)** — Applications of AI in astro- and particle physics
 Summer Term 2026, University of Hamburg, Germany
 2026 **Full Lecture** — Active galactic nuclei and supermassive black holes
 Summer Term 2026, University of Hamburg, Germany
 2025/26 **Seminar** — Extragalactic Astrophysics
 Winter Term 2025/2026, University of Hamburg, Germany
 2024/25 **Full lecture** — Introduction to Astronomy and Astrophysics 2
 Winter Term 2024/2025, University of Hamburg, Germany
 2024 **Full lecture** — Introduction to Astronomy and Astrophysics 1
 Summer Term 2024, University of Hamburg, Germany
 2012–2018 **2 guest lectures** as part of a teaching assistantship
 Steward Observatory, University of Arizona, USA

SCIENTIFIC SOFTWARE DEVELOPMENT

2022 **Lead developer of Sculptor** - Interactive modeling of (electromagnetic) astronomical spectra
 (open-source software, [github](#), [ReadTheDocs](#), [ASCL](#), 9 citations)
 2021 **Co-developer of simqso** - Module for generating simulated quasar spectra
 (open-source software, [github](#), [ReadTheDocs](#), [ASCL](#), 17 citations)
 2020 **Co-developer of Pypelt** - The Python Spectroscopic Data Reduction Pipeline
 (open-source software, [github](#), [ReadTheDocs](#), [Zenodo](#), 191 citations)

TALKS

Invited Talks and Colloquia

- 2026 **LRD Workshop 2026 – Origins, Nature and Cosmic Evolution**, June 2026, [Online workshop](#)
"A little red dot at $z = 7.3$ within a large galaxy overdensity"
- 2026 **Sirius A – Astronomy Symposium**, May 2026, Groningen, Netherlands
"Deciphering the Origins of the First Supermassive Black Holes"
- 2025 **Thüringer Landessternwarte Colloquium**, May 2025, Hamburg, Germany
"The Rise of Supermassive Black Holes at Cosmic Dawn"
- 2025 **Quantum Universe Days (invited plenary)**, February 2025, Hamburg, Germany
"Tracking the Rise of Supermassive Black Holes"
- 2024 **Extragalactic Astrophysics Symposium**, May 2024, Heidelberg, Germany
"The emergence and growth of supermassive black holes in the early Universe"
- 2024 **AEI Seminar**, Feb. 2024, Potsdam, Germany
"The Emergence of Quasars - Growing Supermassive Black Holes in the Early Universe"
- 2023 **Astrophysics & Machine Learning Symposium**, June. 2023, Hamburg, Germany
"Structure evolution in the early Universe: Supermassive Black Holes & Machine Learning"
- 2022 **Colloquium**, Nov. 2022, Hamburg Observatory, Hamburg University
"The Rise and Fall of Quasars"
- 2022 **Königstuhl Colloquium**, Jan. 2022, Heidelberg, Germany
"The Rise and Fall of Quasars - The evolution of quasars across cosmic time"
- 2020 **Early Universe/Reionization Era Conversations at Arizona**, Sep. 2020 Tucson, Arizona, USA
"The X-SHOOTER/ALMA sample and the evolution of reionization-era quasars"
- 2018 **LSST-2018 Project and Community Workshop**, Aug. 2018, Tucson, Arizona, USA
"AGN Selection Opportunities and Challenges with the LSST"
- 2016 **Seismology of the Sun and the Distant Stars**, Jul. 2016, Angra do Heroísmo, Portugal
"Astroseismic Constraints on the Evolutionary Models of Hot Subdwarfs"

Selected Seminar Talks and Conference Contributions since 2020

- 2026 **"Evolution, Near and Far (abridged)"**, Naples, Italy, Jun. 2026
"Pushing the Edge: Euclid Quasar Discoveries at Cosmic Dawn" (remote)
- 2025 **"Multiscale environment of AGN across cosmic time"**, Ringberg castle, Germany, Nov. 2025
"Placing Supermassive BHs in Perspective – Environments of two $z > 7$ Quasars and a $z \sim 7.3$ LRD"
- 2025 **"EREBUS / COSMOS-3D meeting 2025"**, Bologna, Italy, Jun. 2025
"Placing Supermassive Black Holes in Perspective - Environments and Galaxy Clustering around Two Quasars and a Little Red Dot at $z \sim 7.3$ "
- 2024 **"Probing the Genesis of Supermassive Black Holes"**, Kashiwa, Japan, Nov. 2024
"A First Measurement of AGN-Galaxy Clustering at Cosmic Dawn"
- 2024 **MPIA Galaxy Coffee**, Sept. 2024, Max Planck Institute for Astronomy, Heidelberg, Germany
"A galaxy overdensity anchored by a broad-line, low-luminosity AGN at $z = 7.3$ "
- 2024 **"EAS 2024"**, Padova, Italy, Jul. 2024
"Placing a Compact, Red AGN at $z = 7.27$ in Perspective"
- 2023 **"First Light Conference"**, Boston, USA, Jun. 2023
"A first step towards mapping reionisation epoch quasar light-echoes with JWST"
- 2022 **JWST Quasar Workshop**, Ringberg, Oct. 2022
"Towards Tomographic Mapping of Reionization Epoch Quasar Light-Echoes with JWST"
- 2022 **"What drives the growth of black holes: a decade of reflection"**, Reykjavik, Iceland, Sep. 2022
"Supermassive Black Hole Growth in the Reionization Era"
- 2022 **Quasars and Galaxies through Cosmic Time**, virtual conference, Jan. 2022
"Insights into quasar evolution in the first two billion years after the Big Bang"
- 2021 **MPIA Galaxy Coffee**, Feb. 2021, Max Planck Institute for Astronomy, Heidelberg, Germany
"The X-SHOOTER/ALMA sample and the evolution of reionization-era quasars"
- 2020 **Extragalactic group seminar**, Nov. 2020, Institute of Astronomy, Cambridge, UK

- 2020 "The X-SHOOTER/ALMA sample and the evolution of reionization-era quasars"
Summer All Zoom Epoch of Reionization Astronomy Conference, virtual conference, Jul. 2020
 "First Results of the X-SHOOTER/ALMA Sample of Quasars in the Epoch of Reionization"

SERVICE

Service to Profession

- 2026 **Co-Convenor** of two splinter sessions at the German Astronomical Society meeting
 Probing the Central Engine of Active Galactic Nuclei with Variability
 Deciphering AGN from new survey data: eROSITA, 5XMM, Euclid and JWST
- 2024 **Co-lead organizer**, EAS Special Session S1
 From Galaxies to Quasars and back: Linking galaxy and SMBH evolution in the first two billion years
- 2021 **SOC member**, EAS Symposium S7
 Quasars at the reionization epoch
- 2020 **Lead-organizer**, splinter session at German Astronomical Society meeting
 Accreting supermassive black holes through cosmic time
- 2020-2025 **Co-lead** classification working group, LSST AGN science collaboration
 since 2017 **Referee** (MNRAS, A&A, ApJ, PASA, Nature Astronomy, Euclid Consortium, ERC external reviewer, ALMA LP external assessor)

Departmental Service

- 2024 — Work on PhD examination commissions, as a second thesis assessor (Bachelor)
- 2020 — 2021 Seminar organizer (Galaxy coffee, MPIA)
- 2020 — 2021 **Postdoc representative** (MPIA)
- 2019 — 2021 Time allocation committee (MPIA)
- 2019 Ph.D. application committee (IMPRS-HD)
- 2010 — 2011 **Head of physics student representatives** (Kiel)
- 2009 — 2011 Student representative on the board of the department of physics (Kiel)
- 2009 — 2011 Student representative on the examination board of the physics department (Kiel)

PUBLIC OUTREACH

- 2026 **Public Talk** - Sternwarte Lübeck, Lübeck
Das James Webb Weltraumteleskop - Neue Bilder vom Blick zurück ins frühe Universum
- 2026 **Public Talk** - LandFrauen e.V., Eckernförde
Das James Webb Weltraumteleskop - Neue Bilder vom Blick zurück ins frühe Universum
- 2025 **Public Talk** - Sternwarte Hamburg, Hamburg-Bergedorf
Quasare und das Rätsel der supermassereichen Schwarzen Löcher
- 2025 **Public Talk** - Sternwarte Lübeck, Lübeck
Quasare und das Rätsel der supermassereichen Schwarzen Löcher
- 2024 **Public Talk (Live-Stream)** - Astro & Co, Haus der Astronomie, Heidelberg
Auf den Spuren der Quasare
- 2024 **Public Talk** - Sternstundenfestival, Universität Hamburg, Hamburg
Das James Webb Weltraumteleskop - Neue Bilder vom Blick zurück ins frühe Universum
- 2023 **Public Talk** - "Wie alles began ...", Universität Hamburg, Hamburg
Das James Webb Weltraumteleskop
- 2017 **Public Talk** - Astronomy on Tap Series, Borderlands Brewery, Tucson
- 2017 **Invited Article** - "[Warum steht die Erde nie still](#)", German on-line magazine "Krautreporter"
- 2014-2016 **Invited talks** on "Black Holes", Desert Sky Middle School, Tucson

PROFESSIONAL MEMBERSHIPS

Deutsche Physikalische Gesellschaft (German Physical Society)

Astronomische Gesellschaft (German Astronomical Society)

Euclid Consortium (Primeval Universe Science Working Group)

LSST AGN Science Collaboration (Classification subgroup co-lead)

4MOST Consortium (4MOST Science Team)

LOFAR Consortium (Member)